Proposed Solution

Date	1 July
Project Title	Rising Waters - Real-time Flood Prediction and Alert System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform flood management using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and safer communities. Key features include a machine learning-based hydrological predictive model and real-time operational decision-making.

Project Overview	
Objective	The primary objective is to revolutionize the flood monitoring process by implementing advanced machine learning techniques, ensuring faster and more accurate risk assessments.
Scope	The project comprehensively assesses and enhances the disaster response process, incorporating machine learning for a more robust and efficient predictive alert network.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in current manual stream gauge reporting systems adversely affects operational dispatch speed and community evacuation readiness.
Impact	Solving these issues will result in improved operational intelligence, reduced property and life risks, and an overall enhancement in the emergency management response system, contributing to community safety and organizational success.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict precipitation runoffs, creating a dynamic and adaptable hazard monitoring system.
Key Features	• Implementation of a machine learning-based flood risk assessment model. • Real-time decision-making for quicker community emergency alerts. • Continuous learning to adapt to evolving environmental and climate patterns.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU

Resource Type	**Description**	**Specification/Allocation**
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Regional Telemetry Data, 500MB, .csv / OpenWeather API